

the images are not as sharp as those obtained through normal colonoscopy, it is a viable alternative for those who cannot bear the pain or feel embarrassed by the procedure.

Other versions of Pill-Cam help doctors to see the small intestine and oesophagus. The company has also developed SmartPill—an ingestible capsule that measures pressure, pH and temperature as it travels through the gastrointestinal tract. This helps doctors to assess GI motility.

Bravo pH is another capsule-based test that helps to test for acid reflux. The miniature pH capsule attaches to the oesophagus and sends pH data wirelessly to a small recorder worn on a shoulder strap or waistband. Information is collected over multiple days, enabling doctors to study the frequency and duration of acid flowing back up into the oesophagus. This helps to confirm the presence of gastroesophageal reflux disease (GERD). This is a totally catheter-free solution, so the patients can go about normal activities and have a normal diet while the pill unobtrusively monitors their acid reflux. They can remove the receiver to take a shower. This type of monitoring under a normal routine gives better results than keeping the patient under observation.

Unique engineering challenges

The most obvious challenge to edible electronics is, obviously, miniaturisation. It is impossible to implant or consume something that is bulky. With the electronics industry in general moving towards miniaturisation in everything, this goal has become more achievable in implantable devices, too. Unlike



Two types of flexible, biocompatible aqueous sodium-ion batteries developed by a team at Fudan University (Credit: ZhaoWei Guo et al)

the traditional pacemaker, the newer leadless pacemakers are about the size of a large capsule, and can be placed directly in the heart. These can be implanted through a femoral vein puncture in about half an hour. No more bulges or scars, these pacemakers are very unobtrusive and apparently also less prone to infections. The National Heart Centre Singapore (NHCS) began implanting such pacemakers last year.

Thermal management is another huge challenge in implantable devices. Like any other electronic device, implantable devices too vent out waste heat. However, this heat should not harm the surrounding tissues. So while designing an implantable device, engineers have to consider thermal properties of biomaterials, the effect of blood flow on temperature distribution, interfacial contact resistance, effect of temperature change on various types of tissues and communication duty cycles of embedded electronic components.

Further, the in-vivo electronic device must be biocompatible—it must not cause any adverse reaction inside the body. It must be stable over time, despite the temperature and pH variations in the human body. The biocompatible materials must not contain cancer-causing toxins, and must mostly be made of materials naturally suitable to the human body. The insulation materials must match the texture and smoothness of the surface they interface with, be it tissue, bone or skin.

Currently, we have biocompatible polymers for use in drug delivery devices, skin or cartilage and ocular implants;

metals for dental and orthopaedic work; semiconductor materials for bio-sensors and implantable micro-electrodes; and ceramics for bone replacements, heart valves and dental implants. But, ingestible electronics adds to the constraints, requiring the device and its power source to be made of materials that are part of our diet and can be disintegrated and disposed of easily by the body.

The search is on

There is a lot of ongoing research on bio-inspired, biocompatible materials and power sources, which can improve the performance and reliability of implantable and ingestible devices.

A recent paper by MIT researchers, published in *Nature*, describes a biocompatible power source that lasts much longer than current options that can power the edible device only for a few minutes or at the most a few hours. This energy-harvesting galvanic cell for continuous in-vivo temperature sensing and